

Problem 0. n^3 . Yes, matrix multiplication can be done with fewer multiplication operations, by converting them to addition. How much better? Nobody knows. The best result has reduced the number of multiplications to $n^{2.371552}$ as of April 2024

Problem 1.

$$x = r \cos \varphi \quad (1)$$

$$y = r \sin \varphi$$

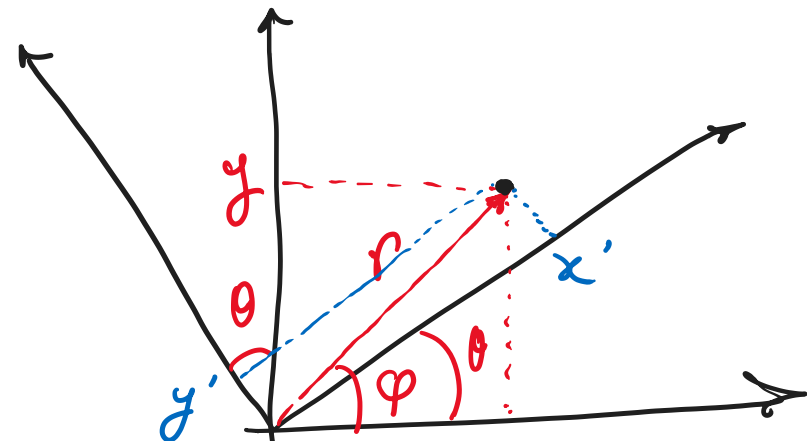
$$x' = r \cos(\varphi - \theta)$$

$$y' = r \sin(\varphi - \theta)$$

$$\text{Recall, } \begin{cases} \cos(\varphi - \theta) = \cos \varphi \cos \theta + \sin \varphi \sin \theta \\ \sin(\varphi - \theta) = \sin \varphi \cos \theta - \cos \varphi \sin \theta \end{cases}$$

$$\Rightarrow \begin{cases} x' = \frac{x}{r \cos \varphi} \cos \theta + \frac{y}{r \sin \varphi} \sin \theta = x \cos \theta + y \sin \theta \\ y' = \frac{y}{r \sin \varphi} \cos \theta - \frac{x}{r \cos \varphi} \sin \theta = -x \sin \theta + y \cos \theta \end{cases}$$

$$\Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

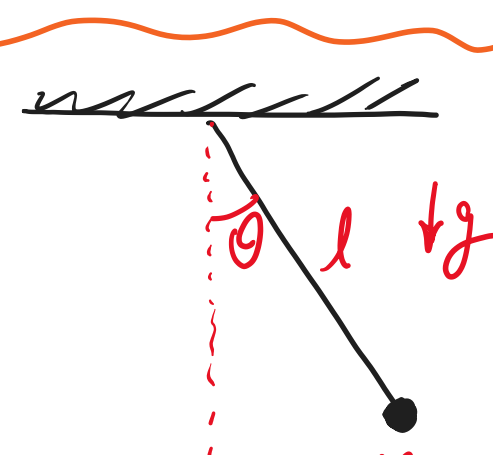


Problem 2. knowns: g : gravity accel.

θ : angle

l : length of pendulum

m : The pendulum mass at its end.



a) unknown: f : frequency

$$[f] = T^{-1} = [g]^\alpha [l]^\beta [m]^\gamma$$

$$= \left(\frac{L}{T^2}\right)^\alpha L^\beta M^\gamma$$

$$\Rightarrow \alpha = +\frac{1}{2}, \beta = -\frac{1}{2}, \gamma = 0$$

$$\Rightarrow f \propto \sqrt{\frac{g}{l}} \quad \text{or} \quad f = f(\theta) \sqrt{\frac{g}{l}}$$

$$b) [Energy] = [Force]^\alpha [distance]^\beta = M L^2 T^{-2}$$

$$= [g]^\alpha [l]^\beta [m]^\gamma = \left(\frac{L}{T^2}\right)^\alpha L^\beta M^\gamma$$

$$M: \gamma = 1$$

$$L: \alpha + \beta = 2$$

$$T: \alpha = 1 \Rightarrow \beta = 1$$

$$\Rightarrow E \propto mgl \quad \text{or} \quad E = f(\theta) mgl$$

How about $f(\theta)$? E must be rotary and lowest at $\theta = 0$.

$\Rightarrow$ Moreover, the potential energy must change proportional to height:

$$h = l(1 - \cos \theta)$$

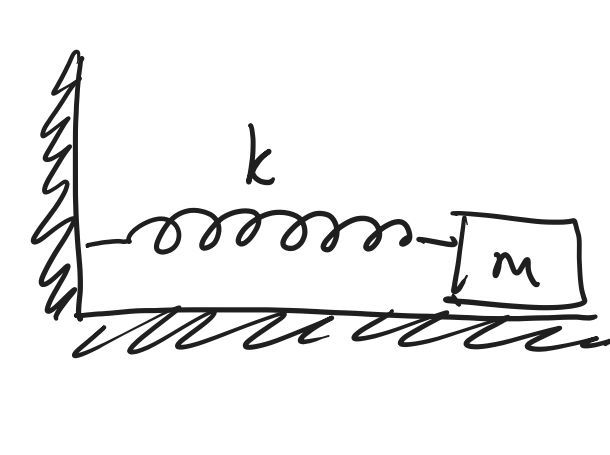
$$\Rightarrow E = mgl(1 - \cos \theta) \quad \text{likely}$$

Problem 3. known: k : spring constant

m : mass attached to spring

x : displacement from equilibrium

equation: $F = ma = -kx$



a) unknown: f : frequency

$$[f] = T^{-1} = [k]^\alpha [m]^\beta [x]^\gamma = \left(\frac{F}{x}\right)^\alpha M^\beta L^\gamma$$

$$= \left(\frac{ML}{T^2L}\right)^\alpha M^\beta L^\gamma$$

$$\Rightarrow \begin{cases} T^{-1} = T^{-2\alpha} \Rightarrow \alpha = \frac{1}{2} \\ \beta = -1 \\ \gamma = 0 \end{cases}$$

$$\Rightarrow f \propto \sqrt{\frac{k}{m}} \quad \text{or} \quad f = c \sqrt{\frac{k}{m}}$$

$$b) [E] = \frac{ML^2}{T^2} = [k]^\alpha [m]^\beta [x]^\gamma = \left(\frac{M}{T^2}\right)^\alpha M^\beta L^\gamma$$

$$\Rightarrow \begin{cases} k: \frac{1}{T^2} = \left(\frac{1}{T^2}\right)^\alpha \Rightarrow \alpha = 1 \\ m: M = M^\alpha M^\beta = M^{\beta+1} \Rightarrow \beta = 0 \\ x: L^2 = L^\gamma \Rightarrow \gamma = 2 \end{cases}$$

$$\Rightarrow E \propto kx^2 \quad \text{or} \quad E = \beta kx^2$$

c) straightforward. Repeat above. $f \propto f\left(x\frac{b}{k}\right) \sqrt{\frac{k}{m}}$

Problem 3. knowns: m : mass of satellite

g : gravity accel.

R : Earth radius

unknowns: v ?

$$[v] = \frac{L}{T} = [m]^\alpha [g]^\beta [R]^\gamma = M^\alpha \left(\frac{L}{T^2}\right)^\beta L^\gamma$$

$$\begin{cases} M: \alpha = 0 \\ L: L^{\beta+\gamma} = L^1 \\ T: T^{-1} = T^{-2\beta} \Rightarrow \beta = \frac{1}{2} \end{cases} \Rightarrow \gamma = \frac{1}{2}$$

$$\Rightarrow v \propto \sqrt{gR} \quad \text{or} \quad v = c \sqrt{gR}$$

Problem 5. knowns: h : height

$\vec{F}_d = -m\alpha v$ drag force

Governing equations:

$$v(t) = -\frac{g}{\alpha} (1 - e^{-\alpha t})$$

$$y(t) = h - \frac{g}{\alpha} \left(t - \frac{1}{\alpha} (1 - e^{-\alpha t})\right)$$

unknown: prove the correctness of eqns. for $t \gg \alpha$ or $t \ll \alpha$.

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \quad \text{Taylor expansion.}$$

$$\text{If } t \ll \alpha \Rightarrow v(t) = -\frac{g}{\alpha} (1 - \exp(-\alpha t)) \approx$$

$$= -\frac{g}{\alpha} \left(1 - [1 - \alpha t + \frac{(\alpha t)^2}{2} - \dots]\right)$$

$$\boxed{v(t) = -gt + O((\alpha t)^2) \quad \text{for } t \ll \alpha}$$

Does this make sense? Yes, because initially drag force is small for v is small, thus acting like free fall.

$$\text{If } t \gg \alpha, y(t) = h - \frac{g}{\alpha} \left(t - \frac{1}{\alpha} (1 - e^{-\alpha t})\right)$$

The rest follows similarly.

$$\text{Problem 6. knowns: } m, M \text{ masses; } v_m = \frac{(m-M)v}{m+M}; v_M = \frac{2mv}{m+M}$$